

Task 11: Session 11 (AIML) - Assignment Questions

Name: Mihir Sopan Patil

Domain: AIML

College & Branch Name: Zeal Polytechnic, Diploma In AIML

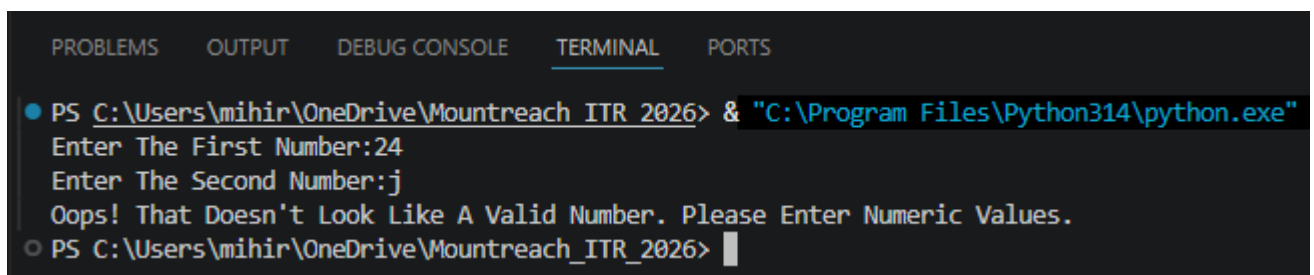
1. Basic try-except

Write a program that takes two numbers as input from the user and prints their division.

Use try-except to handle the case when the user enters non-numeric input (ValueError).

Print a friendly message if an error occurs.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Enter The First Number:24
Enter The Second Number:j
Oops! That Doesn't Look Like A Valid Number. Please Enter Numeric Values.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

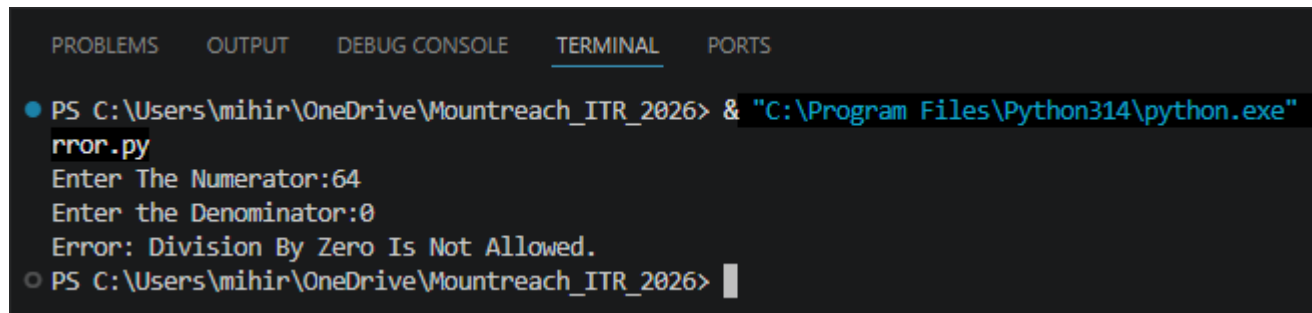
2. Handling ZeroDivisionError

Write a program that divides two numbers entered by the user.

Use try-except to specifically catch ZeroDivisionError and print: "Error: Division by zero is not allowed."

Also handle ValueError for invalid input.

Output:



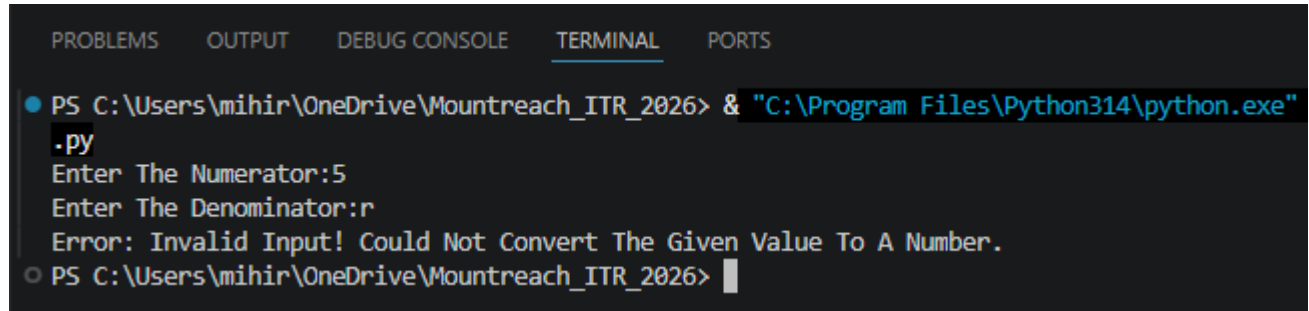
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  error.py
  Enter The Numerator:64
  Enter the Denominator:0
  Error: Division By Zero Is Not Allowed.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

3. Multiple except Blocks

Create a program that performs division and also tries to convert a string to integer.

Handle both ValueError and ZeroDivisionError with separate except blocks and appropriate messages.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
.py
Enter The Numerator:5
Enter The Denominator:r
Error: Invalid Input! Could Not Convert The Given Value To A Number.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> █
```

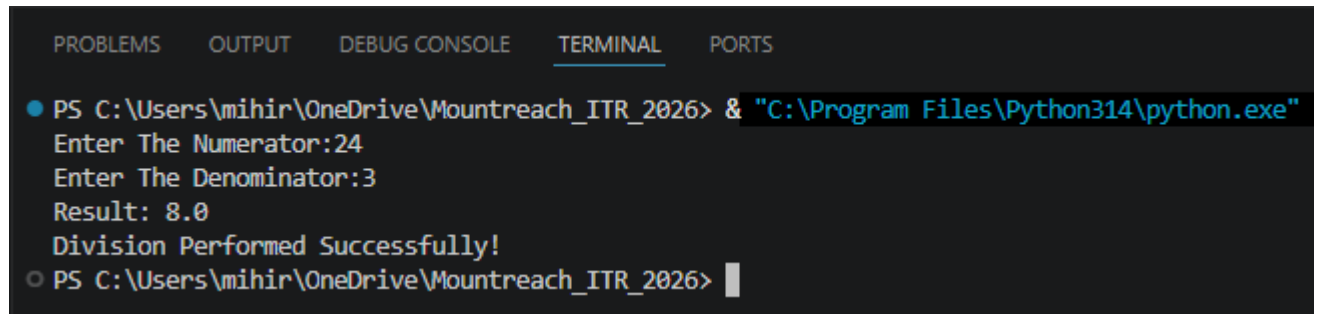
4. Using else

Write a program that takes two numbers and divides them.

Use try, except, and else.

The else block should print "Division performed successfully!" only when no exception occurs.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Numerator:24
Enter The Denominator:3
Result: 8.0
Division Performed Successfully!
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

5. Using finally

Modify the division program to include a finally block that always prints: "Program execution completed."

Demonstrate that finally runs whether an exception occurs or not.

Add comments explaining the purpose of finally.

Output:



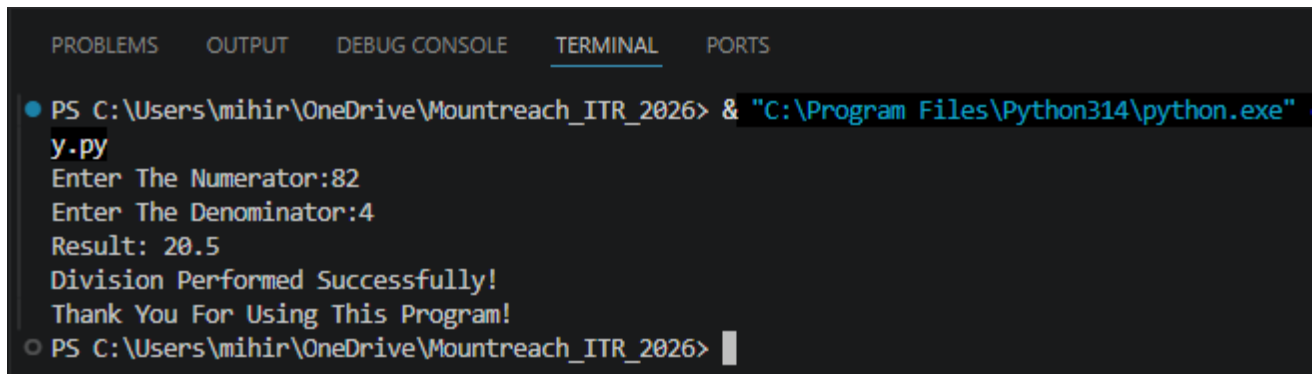
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Numerator:24
Enter The Denominator:6
Result: 4.0
Program Execution Completed.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

6. Combined try-except-else-finally

Write a complete division program using all four blocks (try, except, else, finally):

- Handle ValueError and ZeroDivisionError.
- Print success message in else.
- Always thank the user in finally.

Output:



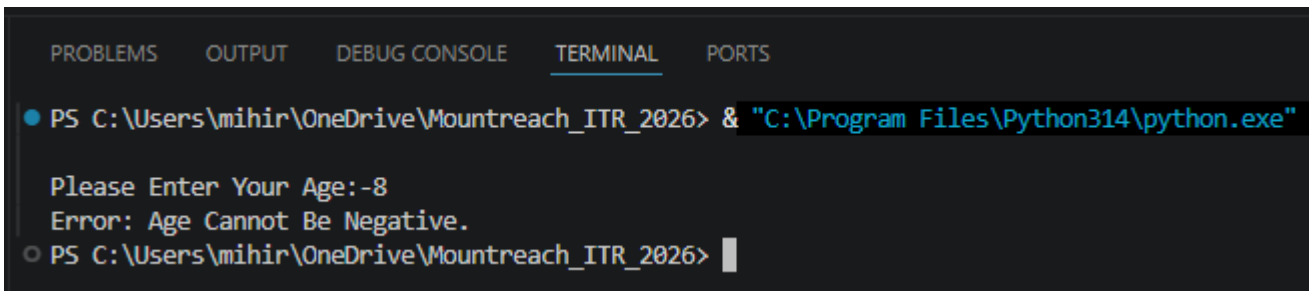
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
y.py
Enter The Numerator:82
Enter The Denominator:4
Result: 20.5
Division Performed Successfully!
Thank You For Using This Program!
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

7. Practical - Age Input

Write a program that asks the user for their age.

- Convert the input to integer.
- If the age is negative, raise a ValueError with message "Age cannot be negative."
- Handle the exception gracefully and print a suitable message.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Please Enter Your Age:-8
Error: Age Cannot Be Negative.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

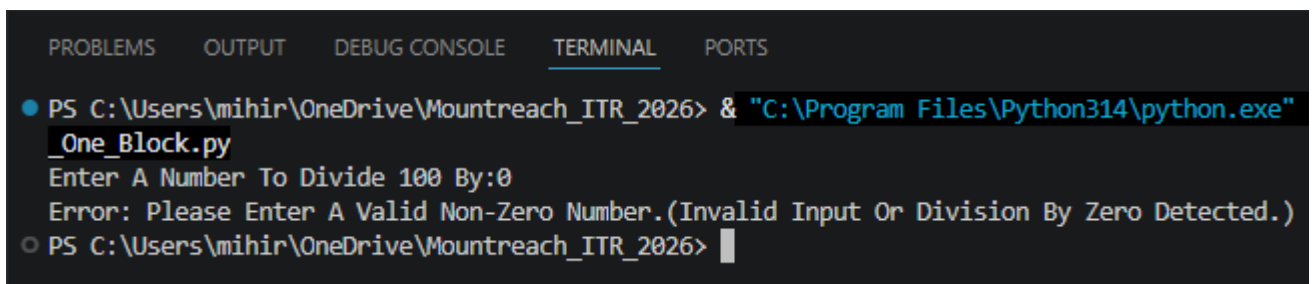
8. Multiple Exceptions In One Block

Create a program that:

- Takes a number as input.
- Divides 100 by that number.
- Handle both `ValueError` and `ZeroDivisionError` in a single except block using a tuple.

Print a combined error message.

Output:



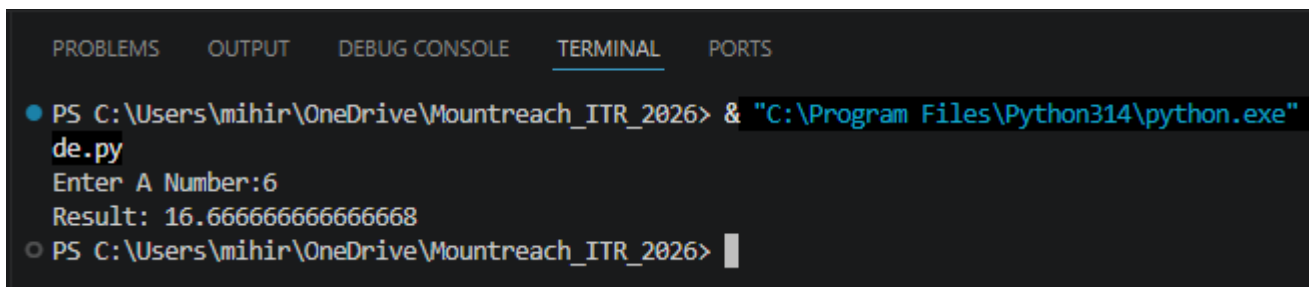
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  _One_Block.py
  Enter A Number To Divide 100 By:0
  Error: Please Enter A Valid Non-Zero Number.(Invalid Input Or Division By Zero Detected.)
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```


9. Debugging Exception Code

The following code has issues. Correct it and add comments explaining the fixes:

```
try
    num = int(input("Enter a number: "))
    result = 100 / num
    print("Result:", result)
except ValueError
    print("Please enter a valid number")
except:
    print("Some error occurred")
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
de.py
Enter A Number:6
Result: 16.666666666666668
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

10. Mini Project - Exception Handling

Create a **Simple Calculator** with exception handling:

The program should run in a loop and show this menu:

- a) Addition
 - b) Subtraction
 - c) Multiplication
 - d) Division
 - e) Exit
- Take user choice and two numbers.
 - Perform the operation inside try block.
 - Handle ValueError (invalid input) and ZeroDivisionError (for division).
 - Use finally to print "Operation attempted." after every calculation.
 - Continue until user chooses to exit.

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

**** Calculator Menu ****
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter Your Choice (1-5): 1
Enter The First Number:g
Error: Invalid Input! Please Enter Numeric Values.
Operation Attempted.

**** Calculator Menu ****
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter Your Choice (1-5): 4
Enter The First Number:3
Enter The Second Number:0
Error: Division By Zero Is Not Allowed!
Operation Attempted.

**** Calculator Menu ****
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter Your Choice (1-5): 5
Exiting The Calculator. Goodbye!
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```